QUENCH.

QUANTIFIED EVIDENCE NAVIGATOR FOR COMBUSTION HAZARDS

Team [TEAM NAME] · Members [NAMES OF TEAM MEMBERS]

Which flight tests match the **oxygen and pressure** of a proposed lunar cabin?

24
PSI investigations catalogued

1,156
distinct flight test records in the CSV

34 %
proposed exploration atmosphere at 8.2 psia

15
gas tests within the project's comparison window

01

Decades of fire experiments in orbit. Do they cover a proposed Moon-base atmosphere?

The challenge: a dashboard that **summarizes, ranks and interprets** NASA's microgravity combustion investigations for fire safety.

1,156 distinct flight test records from nine of 24 catalogued investigations. The map uses ambient oxygen and chamber pressure; oxygen supplied to a burner is kept separate.

> "The intermediate oxygen concentration used on the Moon will likely be 30% or greater, with a corresponding pressure of 10.2 psi or less."
>
> Flame in Freefall challenge statement

Both conditions match **20 gas tests** in our tables, with no matching droplet or solid tests. NASA's separate **Saffire-VI** flight results include solids in this region.

179

Normal-flame records
Ambient oxygen is reported.

154

Inverse-flame records
Oxygen is measured in the burner feed.

15

Normal-flame gas tests within ±4 points O_2
and ±15% pressure of 34% / 8.2 psia.

02

Rank: which investigations matter for your habitat

RANK	INVESTIGATION	WHAT BURNED	SOURCE ROWS	AMBIENT O_2 %	NEAR THIS CABIN	SCORE	COMPARE
01	PSI-99 **SAFFIRE-II** PREVENTION Nine sample entries compare orbital and normal-gravity results for silicone, cotton–fiberglass fabric, PMMA and Nomex. All four silicone entries record approximately zero orbital burn length. The entry combining PMMA and Nomex specifies that only the PMMA burned.	Solid materials	15	21.5 – 22.1	Recorded ambient oxygen 21.5–22.1%; no values in the cabin window	82	☐
02	PSI-98 **SAFFIRE-I** PREVENTION A large cotton–fiberglass sheet burned inside an uncrewed Cygnus after it left the ISS. The table gives dimensions of 94 by 40.6 cm and lists ignition power, airflow and burn duration.	Solid materials	4	21.5 – 21.7	Recorded ambient oxygen 21.5–21.7%; no values in the cabin window	73	☐
03	PSI-69 **FLEX** SUPPRESSION Methanol and heptane droplets burned in many atmospheres to find when a flame cannot survive, and how well carbon dioxide and helium put it out. 274 runs with recorded outcomes, from 12 to 34 % oxygen.	Liquid droplets	274	12 – 34	24 runs in the ambient oxygen window (24 at another pressure)	72	☐
04	PSI-25 **BASS-II** PREVENTION BASS-II tested PMMA and other solid materials in a flow duct aboard the ISS. Initial oxygen in the imported table ranged from 13.6 to 22.2 %. The table records conditions and gas readings. Images, video and the investigators' report document how individual samples burned.	Solid materials	129	13.6 – 22.2	Recorded ambient oxygen 13.6–22.2%; no values in the cabin window	67	☐
05	PSI-159	Gas flames	232	18 – 40	15 runs inside this cabin's	59	☐

- Pick a **station, spacecraft, Moon base or rover, or Mars base**. All 24 investigations are ranked for it.
- The score has four visible parts: named for fire safety by NASA, fuel close to cabin materials, tested near that cabin's air, data ready to use.
- You can change the weights. Every rule is printed.
- `PREVENTION` `DETECTION` `SUPPRESSION` follow the challenge statement word for word.

03

Summarize and compare: what each one burned, varied and measured

SIDE BY SIDE The same in all of them: how the challenge statement treats it. Rows that match are marked.

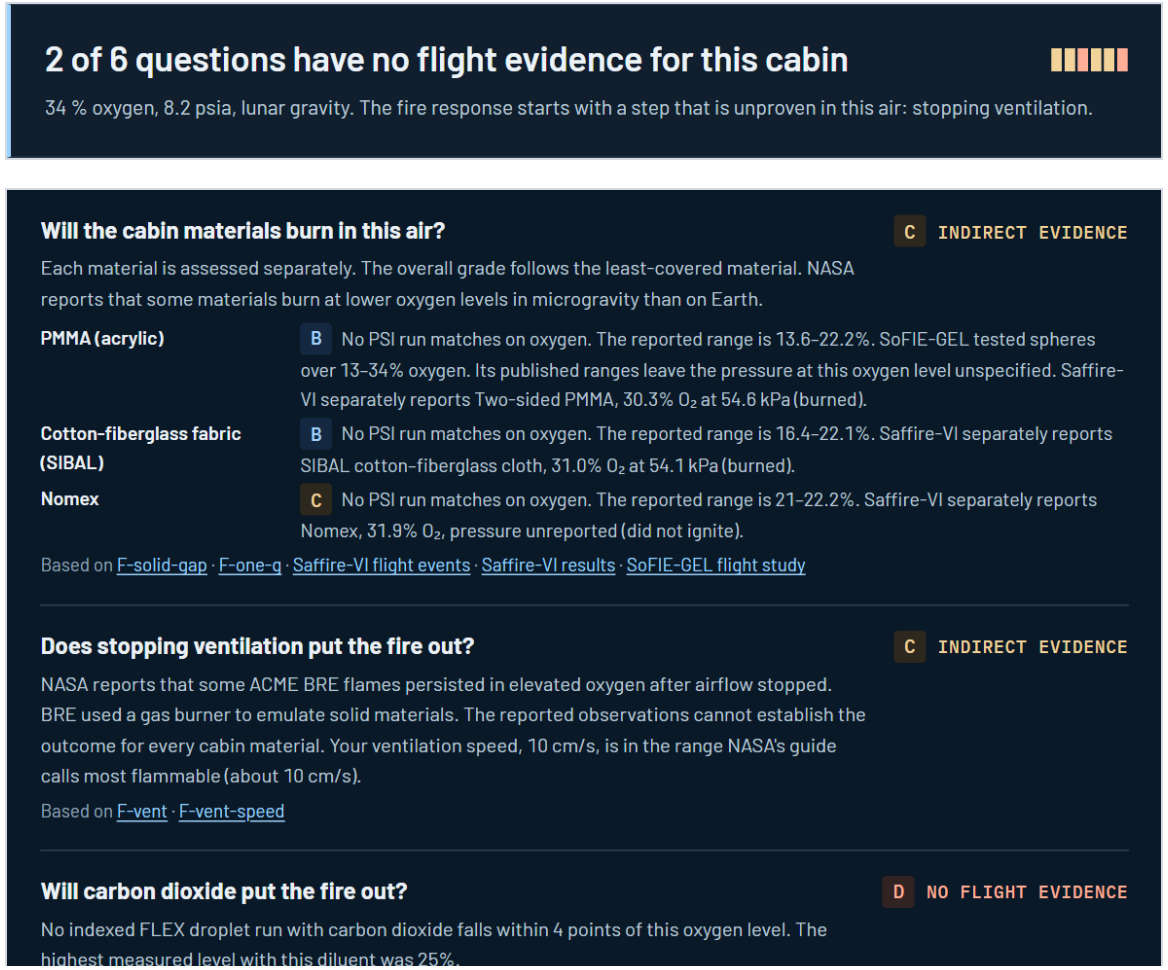
Clear

	BASS-II	SAME	FLEX
ROLE IN FIRE SAFETY	Prevention	Detection	Suppression
HOW THE CHALLENGE STATEMENT TREATS IT SAME	Named in the challenge statement as a fire-safety study	Named in the challenge statement as a fire-safety study	Named in the challenge statement as a fire-safety study
WHAT BURNED	Solid materials	Smoke from overheated materials	Liquid droplets
FUELS IN THE TABLE	2 cm SIBAL; 1/4" clear rod; 1/4" black rod; 1 cm SIBAL; and 45 more	Teflon; Kapton: same conditions as 1-4; Silicone Rubber: same conditions as 1-4; Lampwick: same conditions as 1-4; and 1 more	Methanol; Heptane
SAMPLE OR BURNER SIZE	–	–	Droplet initial diameter; mm: 1.1 to 4.67 (163 values)
HARDWARE	Microgravity Science Glovebox, SPICE flow duct	Smoke and aerosol instruments	Droplet apparatus (MDCA) in the Combustion Integrated Rack
PLATFORM AND YEARS	Space station, 2010 to 2022	Space station, 2001 to 2008	Space station, 2007 to 2016
AMBIENT OXYGEN, %	13.6 – 22.2	–	12 – 34
BURNER OXIDIZER OXYGEN, %	–	–	–

- Each investigation opens to its objective, fuels with sample or burner size, **what changed between runs, what stayed the same** and what was recorded.
- These lists are computed from NASA's run tables, not typed in.
- Tick two or three to see them **side by side**. Matching rows are marked.
- Six hardware families help identify shared apparatus. Fuel, flow and atmosphere can still differ.

04

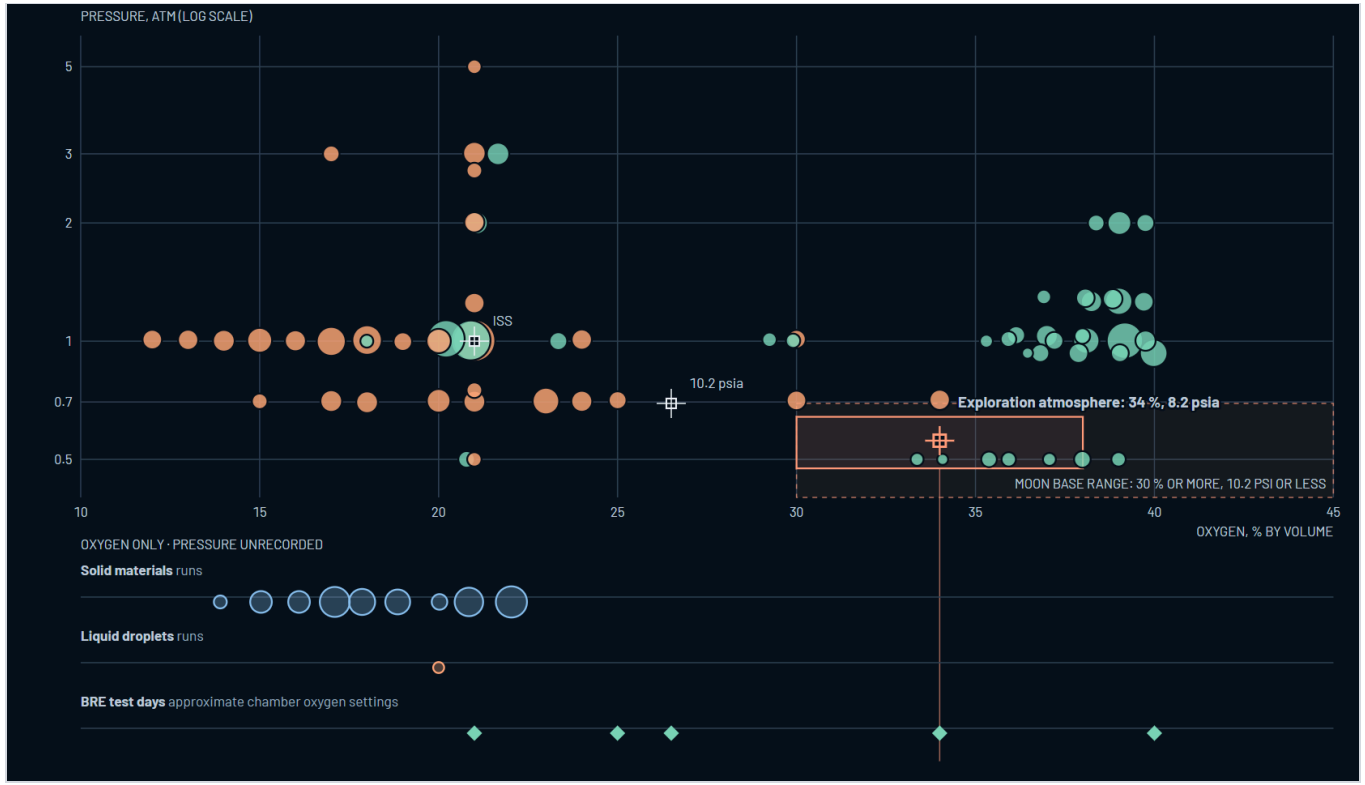
Interpret: grade the evidence for any cabin and list the tests still needed



- **Mission check:** set oxygen, pressure, gravity, ventilation, materials and extinguishing agent. Six fire-safety questions are graded A to D by written rules.
- The scorecard separates **two-coordinate matches** from tests with missing pressure, and flags results outside its tables.
- **15 ranked findings** for prevention, detection and suppression, each with evidence, limits, an action and NASA sources.
- In preliminary BRE gas-burner experiments, some flames persisted after forced airflow stopped.

05

Data for models: every run as CSV, a predictor that knows its limits, results read from NASA video



82 %

right on test days the model never saw

69 %

always guessing the common outcome

- All **1,156 records download as CSV**, or only those near your cabin.
- Predictor trained on 261 FLEX droplet runs. It flags extrapolation.
- BASS-II video: one calibration run and **five** consistency checks.

06

Built to be trusted, free to run, open about SI

TRUST

- The run count can be rebuilt from the archived tables.
- Quotations are checked against archived source text.
- Regression tests check the two CFI-G configurations, missing values and matching windows.

USE OF SI

- SI (Claude) helped write code and draft summaries, findings and answers.
- Code checks cited records and numeric tokens; these checks cannot establish scientific validity.
- No model runs when you use it: no API, no key, no cost.

NEXT

- Extend exports with run-level Saffire and FLEX-2 measurements.
- Publish more measurements extracted from BASS-II video.
- On Earth the same science already made firefighter gear and smoke detectors better.

NASA data: Physical Sciences Informatics combustion investigations, BASS-II raw video and report, Researcher's Guide to ISS Combustion Science. Research prototype, not a design basis. Not an official NASA product. Slides laid out with SI assistance; pictures are screenshots of the live site.